Analysis of Two-Mode Networks Using R

1. Introduction

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Outline

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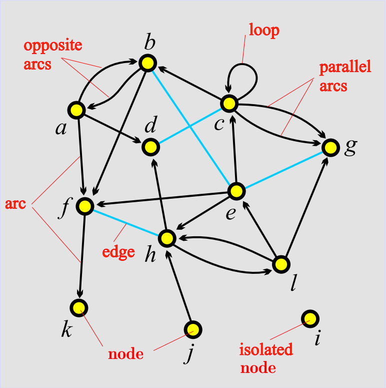
Current version of slides (August 13, 2026 at 15 : 05): slides PDF
https://github.com/bavla/Nets/blob/master/ws/ws26.md

Graph

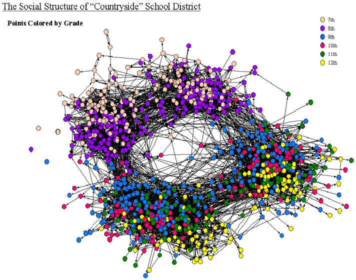


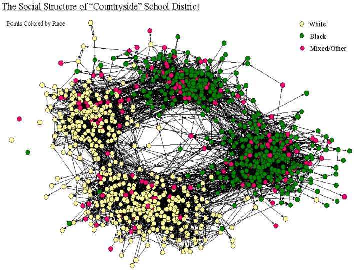
unit, actor – *node*, vertex
tie, line – *link*, edge, arc

arc = directed link, (a, d)
a is the *initial* node,
d is the *terminal* node.

edge = undirected link, $(c : d)$
c and d are *end* nodes.

Display of properties – school (Moody)





Besides the graph, we need data to understand the network!

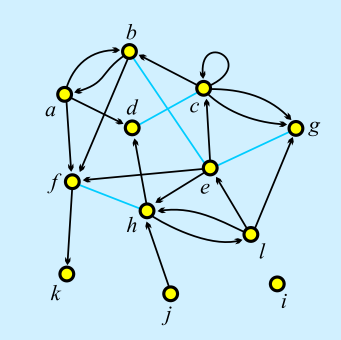
Networks / Formally

Network $=$ Graph $+$ Data

A *network* $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$ consists of:

- a *graph* $\mathcal{G} = (\mathcal{V}, \mathcal{L})$, where $\mathcal{V}$ is the set of nodes, $\mathcal{A}$ is the set of arcs, $\mathcal{E}$ is the set of edges, and $\mathcal{L} = \mathcal{E} \cup \mathcal{A}$ is the set of links.
  $n = |\mathcal{V}|$, $m = |\mathcal{L}|$
- $\mathcal{P}$ *node value functions* / properties: $p : \mathcal{V} \to A$
- $\mathcal{W}$ *link value functions* / weights: $w : \mathcal{L} \to B$

Graph / Sets



$$\mathcal{V} = \{a, b, c, d, e, f, g, h, i, j, k, l\}$$

$$\mathcal{A} = \{(a, b), (a, d), (a, f), (b, a), (b, f), (c, b), (c, c), (c, g)_1, (c, g)_2, (e, c), (e, f), (e, h), (f, k), (h, d), (h, l), (j, h), (l, e), (l, g), (l, h)\}$$

$$\mathcal{E} = \{(b: e), (c: d), (e: g), (f: h)\}$$

$$\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{E})$$

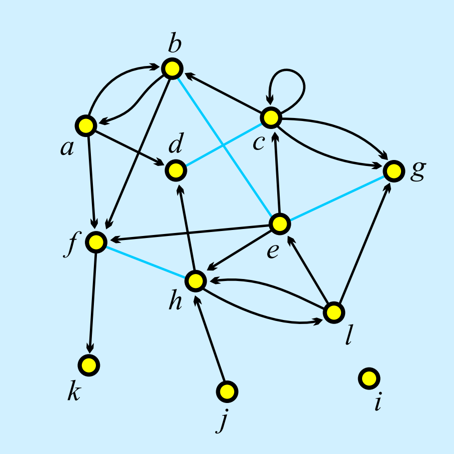
$$\mathcal{L} = \mathcal{A} \cup \mathcal{E}$$

$\mathcal{A} = \emptyset$ – *undirected* graph; $\mathcal{E} = \emptyset$ – *directed* graph.

The R package igraph **doesn't support** the mixing of arcs and edges. A graph is either directed or undirected.

In most applications, an edge $(u: v)$ can be replaced by a pair of opposite arcs (u, v) and (v, u).

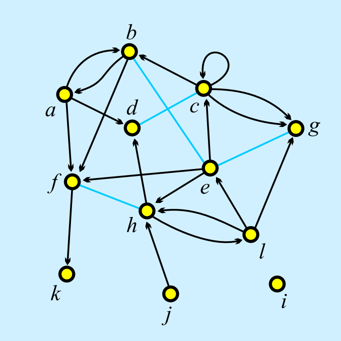
Graph / Sets – NET




```
*Vertices  12
  1 "a"     0.1020   0.3226
  2 "b"     0.2860   0.0876
  3 "c"     0.5322   0.2304
  4 "d"     0.3259   0.3917
  5 "e"     0.5543   0.4770
  6 "f"     0.1552   0.6406
  7 "g"     0.8293   0.3249
  8 "h"     0.4479   0.6866
  9 "i"     0.8204   0.8203
 10 "j"     0.4789   0.9055
 11 "k"     0.1175   0.9032
 12 "l"     0.7095   0.6475
*Arcs
  1   2
  2   1
  1   4
  1   6
  2   6
  3   2
  3   3
  3   7
  3   7
  5   3
  5   6
  5   8
  6  11
  8   4
 10   8
 12   5
 12   7
  8  12
 12   8
*Edges
  2   5
  3   4
  5   7
  6   8
```

factorization !!!

Graph / Matrix



	a	b	c	d	e	f	g	h	i	j	k	l
a	0	1	0	1	0	1	0	0	0	0	0	0
b	1	0	0	0	1	1	0	0	0	0	0	0
c	0	1	1	1	0	0	2	0	0	0	0	0
d	0	0	1	0	0	0	0	0	0	0	0	0
e	0	1	1	0	0	1	1	1	0	0	0	0
f	0	0	0	0	0	0	0	1	0	0	1	0
g	0	0	0	0	1	0	0	0	0	0	0	0
h	0	0	0	1	0	1	0	0	0	0	0	1
i	0	0	0	0	0	0	0	0	0	0	0	0
j	0	0	0	0	0	0	0	1	0	0	0	0
k	0	0	0	0	0	0	0	0	0	0	0	0
l	0	0	0	0	1	0	1	1	0	0	0	0

The graph G is *simple* if in the corresponding matrix all entries are 0 or 1.

Matrix representation provides a link to the tools of linear algebra.

Networks in igraph

The network $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$ can be described by two tables $(\mathcal{V}, \mathcal{P})$ and $(\mathcal{L}, \mathcal{W})$.

GraphSet.net, nodes.csv, links.csv

```
> nWdir <- paste0("https://raw.githubusercontent.com/",
+  "bavla/Nets/refs/heads/master/netsWeight/")
> (V <- read.csv(paste0(nWdir,"data/nodes.csv"),sep=""))
> (L <- read.csv(paste0(nWdir,"data/links.csv"),sep=""))
> (N <- graph_from_data_frame(L,directed=TRUE,vertices=V))
> V(N)$name <- V(N)$label
> N <- delete_vertex_attr(N,"label")
> E(N)$weight <- sample(1:7,ecount(N),replace=TRUE)
> N
> plot(N,edge.width=E(N)$weight)
> N$name <- "Network from data frame example"
> N$by <- "Vladimir Batagelj"
> N$cdate <- date()
> N
> saveRDS(N,file="igraphDF.rds")

> graph_attr(N)
> (nodes <- as_data_frame(N,what="vertices"))
> (links <- as_data_frame(N,what="edges"))
```

Types of networks

Besides ordinary (directed, undirected, mixed) networks, some special types of networks are also used:

- *2-mode networks*, bipartite (valued) graphs – networks between two disjoint sets of nodes.
- *multi-relational networks*.
- *linked networks* and *collections of networks*.
- *multilevel networks*.
- *temporal networks*, dynamic graphs – networks changing over time.
- specialized networks: representation of genealogies as *p-graphs*; *Petri's nets*, molecular graphs, etc.

Network (input) file formats should provide the means of expressing all of these types of networks. All interesting data should be recorded (respecting privacy).

Types of networks

In a *two-mode* network $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, \mathcal{P}, \mathcal{W})$ the set of nodes consists of two disjoint sets of nodes $\mathcal{U}$ and $\mathcal{V}$, and all the links from $\mathcal{L}$ have one end node in $\mathcal{U}$ and the other in $\mathcal{V}$.

A *multi-relational network* $\mathcal{N} = (\mathcal{V}, (\mathcal{L}_1, \mathcal{L}_2, \ldots, \mathcal{L}_k), \mathcal{P}, \mathcal{W})$ contains different relations $\mathcal{L}_i$ (sets of links) over the same set of nodes. Also, the weights from $\mathcal{W}$ are defined on different relations or their union.

In a *linked* or *multimodal* network $\mathcal{N} = ((\mathcal{V}_1, \mathcal{V}_2, \ldots, \mathcal{V}_j), (\mathcal{L}_1, \mathcal{L}_2, \ldots, \mathcal{L}_k), \mathcal{P}, \mathcal{W})$ the set of nodes $\mathcal{V}$ is partitioned into subsets (*modes*) $\mathcal{V}_i$, $\mathcal{L}_s \subseteq \mathcal{V}_p \times \mathcal{V}_q$, and properties and weights are usually partial functions.

Types of networks

A set of networks $\{\mathcal{N}_1, \mathcal{N}_2, \ldots, \mathcal{N}_k\}$ in which each network shares a (sub)set of nodes with some other network is called a *collection* of networks.

A linked network can be transformed into a collection of networks and vice versa.

Bibliographical information is usually represented as a collection of bibliographical networks $\{\mathbf{Ci}, \mathbf{WA}, \mathbf{WK}, \mathbf{WC}, \mathbf{WI}, \ldots\}$ (W – works, A – authors, K – keywords, C – countries, I – institutions) [4].

Another example of multimodal multirelational networks are knowledge graphs. They can have a very diverse structure (a large number of types of units (modes) and predicates (relations)), which allows for a fairly accurate description of facts from a selected field and solving problems about it.

netsJSON

For archiving networks, we can use the JSON format supported by the package `netsWeight` [6, 7].

```
{
"netsJSON": "basic",
"info": {
   "org":1, "nNodes":n, "nArcs":mA, "nEdges":mE,
   "simple":TF, "directed":TF, "multirel":TF, "mode":m,
   "network":fName, "title":title,
   "time": { "Tmin":tm, "Tmax":tM, "Tlabs": {labs} },
   "meta": [events], ...
   },
"nodes": [
   { "id":nodeId, "lab":label, "x":x, "y":y, ...},
   ***
   ],
"links": [
   { "type":arc/edge, "n1":nodeID1, "n2":nodeID2, "rel":r, ...},
   ***
   ],
"data": {
   "data1":description1,
   ***
   }
}
```

netsJSON

```
> saveRDS(AW,file="AW.rds")
> AW1 <- readRDS(file="AW.rds")
> AW1
> write_graph_netsJSON(AW,file="AW.json")
> AW2 <- netsJSON_to_graph(fromJSON("AW.json"),directed=TRUE)
> AW2
> AW3 <- netsJSON_to_graph(fromJSON("AW.json"),directed=FALSE)
> AW3
> V(AW3)[[]]
> E(AW3)[[]]
> graph_attr(AW3)
```

We can combine individual networks into a collection

```
> Bib <- list(netsJSON="Collection",
+ info=list(date=date(),by="VB"),nets=list(AW=AW,WK=WK))
> str(Bib)
> saveRDS(Bib,file="BibNets.rds")
```

Properties (attributes)

Properties of nodes $\mathcal{P}$ and links $\mathcal{W}$ (weights) can be measured on different scales: *numerical* (age, weight, number of contacts), *ordinal* (level of education), and *nominal/categorical* (citizenship, sex, type of contact). They can be *input* as data or *computed* from the network.

Recently, properties with *structured* values are also considered: intervals, lists of keywords, event sequences, distributions, temporal quantities, etc.

We also have to specify how to compute using the values of each property. Usually, for a numerical property $p : \mathcal{V} \to \mathbb{R}$, the real number addition $+$ is used. In a general case $p : \mathcal{V} \to S$, we assume that $(S, +, 0)$ is an *Abelian monoid*; and in the case of a weight $w : \mathcal{L} \to S$, $(S, +, \cdot, 0, 1)$ is a *semiring* [1].

Example of semiring

To construct a semiring corresponding to the balance problem in signed networks ($w : \mathcal{L} \to \{n, p\}$, n – negative, p – positive) we take the set $S = \{0, n, p, a\}$ with four elements (Doreian, 1970; Harary et al., 1965):

- 0 – no walk;
- n – all walks are negative;
- p – all walks are positive;
- a – at least one positive and at least one negative walk.

Balance Semiring

$+$	0	n	p	a
0	0	n	p	a
n	n	n	a	a
p	p	a	p	a
a	a	a	a	a

$\cdot$	0	n	p	a
0	0	0	0	0
n	0	p	n	a
p	0	n	p	a
a	0	a	a	a

x	$x^\star$
0	p
n	a
p	p
a	a

Properties

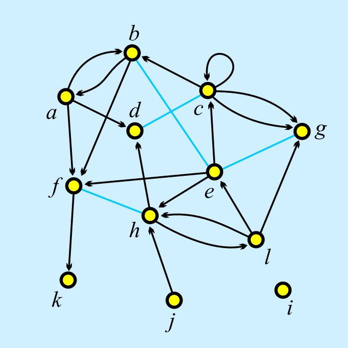
Traditional measurements can be represented (using factorization) as a numerical vector $\mathbf{v} = [v_i]$.

`Clustering` – partition of elements (nodes/links) – *nominal* or *ordinal* data about elements
$v_i \in \mathbb{N}$: element i belongs to the cluster/group v_i;

`Vector` – *numeric* data about elements
$v_i \in \mathbb{R}$: the property has value v_i on element i;

`Permutation` – *ordering* of elements
$v_i \in \mathbb{N}$: element i is at the v_i-th position.

Degrees



degree of node v, $\deg(v)$ = number of links with v as an endnode – a *star*;
indegree of node v, $\text{ideg}(v)$ = number of links with v as a terminal node – an *instar*. An edge endnode is both initial and terminal;
outdegree of node v, $\text{odeg}(v)$ = number of links with v as an initial node – an *outstar*.
For a graph,
initial node $v \Leftrightarrow \text{ideg}(v) = 0$
terminal node $v \Leftrightarrow \text{odeg}(v) = 0$

$$n = 12,\ m = 23,\ \text{ideg}(e) = 3,\ \text{odeg}(e) = 5,\ \deg(e) = 6$$

$$\sum_{v\in\mathcal{V}} \text{ideg}(v) = \sum_{v\in\mathcal{V}} \text{odeg}(v) = |\mathcal{A}| + 2|\mathcal{E}| - |\mathcal{E}_0|,\ \sum_{v\in\mathcal{V}} \deg(v) = 2|\mathcal{L}| - |\mathcal{L}_0|$$

Weighted degrees

weighted in/out-degree

$$\mathrm{wd}(v) = \sum_{e \in \mathrm{star}(v)} w(e)$$

$$\mathrm{wid}(v) = \sum_{e \in \mathrm{istar}(v)} w(e)$$

$$\mathrm{wod}(v) = \sum_{e \in \mathrm{ostar}(v)} w(e)$$

$$\mathrm{wto}(v, u) = \sum_{e \in \mathrm{ostar}(v) \cap \mathrm{istar}(u)} w(e)$$

Neighbors

$N(v)$ is the set of *neighbors* of node v – other endnodes of links containing the node v.

$iN(v)$ is the set of *in-neighbors* of node v – other nodes from which a link leads to the node v.

$oN(v)$ is the set of *out-neighbors* of node v – other nodes to which a link leads from the node v.

Important nodes and links in network

To identify important/interesting elements (nodes, links) in a network, we often try to express our intuition about important/interesting elements using an appropriate measure (index, weight) following the scheme

> *The larger is the measured value of an element, the more important/interesting this element is*

Too often, in the analysis of networks, researchers uncritically pick some measures from the literature. For a formal approach, see Roberts.

The (importance) measure can be obtained as input data – an *observed* property, or computed from the network description – a *structural* property.

An interesting question is studying associations among structural and observed properties. Can an observed property be explained with some structural property/ies?

. . . Important nodes and links in network

The most important distinction between different node *indices* is based on the view/decision of whether the network is considered directed or undirected – closeness, betweenness, clustering, eigen-vector, . . .

Using node or link cuts, islands, cores, skeletons (spanning tree, Pathfinder, k-neighbors), community detection, hubs and authorities, etc., we can identify the most active subnetworks. We can also apply clustering and blockmodeling methods [3].

Cuts

The *node-cut* of a network $\mathcal{N} = (\mathcal{V}, \mathcal{L}, p)$, $p : \mathcal{V} \to \mathbb{R}$, at selected level t is a subnetwork $\mathcal{N}(t) = (\mathcal{V}', \mathcal{L}(\mathcal{V}'), p)$, determined by the set

$$\mathcal{V}' = \{v \in \mathcal{V} : p(v) \geq t\}$$

and $\mathcal{L}(\mathcal{V}')$ is the set of links from $\mathcal{L}$ that have both endnodes in $\mathcal{V}'$.

The *link-cut* of a network $\mathcal{N} = (\mathcal{V}, \mathcal{L}, w)$, $w : \mathcal{L} \to \mathbb{R}$, at selected level t is a subnetwork $\mathcal{N}(t) = (\mathcal{V}(\mathcal{L}'), \mathcal{L}', w)$, determined by the set

$$\mathcal{L}' = \{e \in \mathcal{L} : w(e) \geq t\}$$

and $\mathcal{V}(\mathcal{L}')$ is the set of all endnodes of the links from $\mathcal{L}'$.

netsWeight

We started to develop an R package `netsWeight` over `igraph`, extending its capabilities with some tools for analysis of weighted networks. The package `netsWeight` contains functions

`interlinks(N,atn,c1,c2,col1="red",col2="blue")` – extracts from a given network N, a subnetwork determined by node clusters C_1 and C_2, and links between them. C_i is the set of nodes having for the attribute with name *atn* the value c_i. Its nodes have color col_i.

`node_cut(N,atn,t)` – node cut in N for the node property *atn* at threshold t.

`link_cut(N,atn,t)` – link cut in N for the link property *atn* at threshold t.

The threshold value t is determined based on the distribution of values of weight w or property p. Usually, we are interested in cuts that are not too large, but also not trivial.

Weighted network matrix

A two mode network $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, w)$ can be represented with the corresponding *matrix* $\mathbf{N} = [n[u, v]]_{u \in \mathcal{U}, v \in \mathcal{V}}$ defined as

$$n[u, v] = \begin{cases} \text{wto}(u, v) & (u, v) \in \mathcal{L} \\ \square & \text{otherwise} \end{cases}$$

We represent the number 0 with two symbols, 0 (weight 0) and $\square$ (no link) where $\square = 0$ with rules $\square + a = a$ and $\square \cdot a = \square$.

In the case $\mathcal{U} = \mathcal{V}$ we obtain a square matrix of an ordinary network.

When collecting the network data, consider providing as many properties as possible.

Binarization and transposition

A *binarization* bin($\mathbf{N}$) of the network matrix $\mathbf{N}$

$$\mathrm{bin}(\mathbf{N})[u,v] = \begin{cases} 1 & n[u,v] \neq \square \\ \square & n[u,v] = \square \end{cases}$$

The *transpose* $\mathbf{N}^T$ of matrix $\mathbf{N}$ is a matrix on $\mathcal{V} \times \mathcal{U}$ with entries $n^T[u,v] = n[v,u]$.

$\mathbf{VU} = \mathbf{UV}^T$ and $uv^T[u,v] = vu[u,v]$.

The corresponding network $\widetilde{\mathcal{N}} = ((\mathcal{V},\mathcal{U}),\widetilde{\mathcal{L}},\tilde{n})$ is called the *reversed* network. The direction of its arcs is reversed $\widetilde{\mathcal{L}} = \{(v,u) : (u,v) \in \mathcal{L}\}$, but the weights remain unchanged $\tilde{n}(v,u) = n(u,v)$ for $(u,v) \in \mathcal{L}$.

$$\mathrm{wod}_{\widetilde{\mathcal{N}}}(v) = \mathrm{wid}_{\mathcal{N}}(v), \ldots$$

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References III

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- https://github.com/bavla/NormNet/tree/main/TwoMode